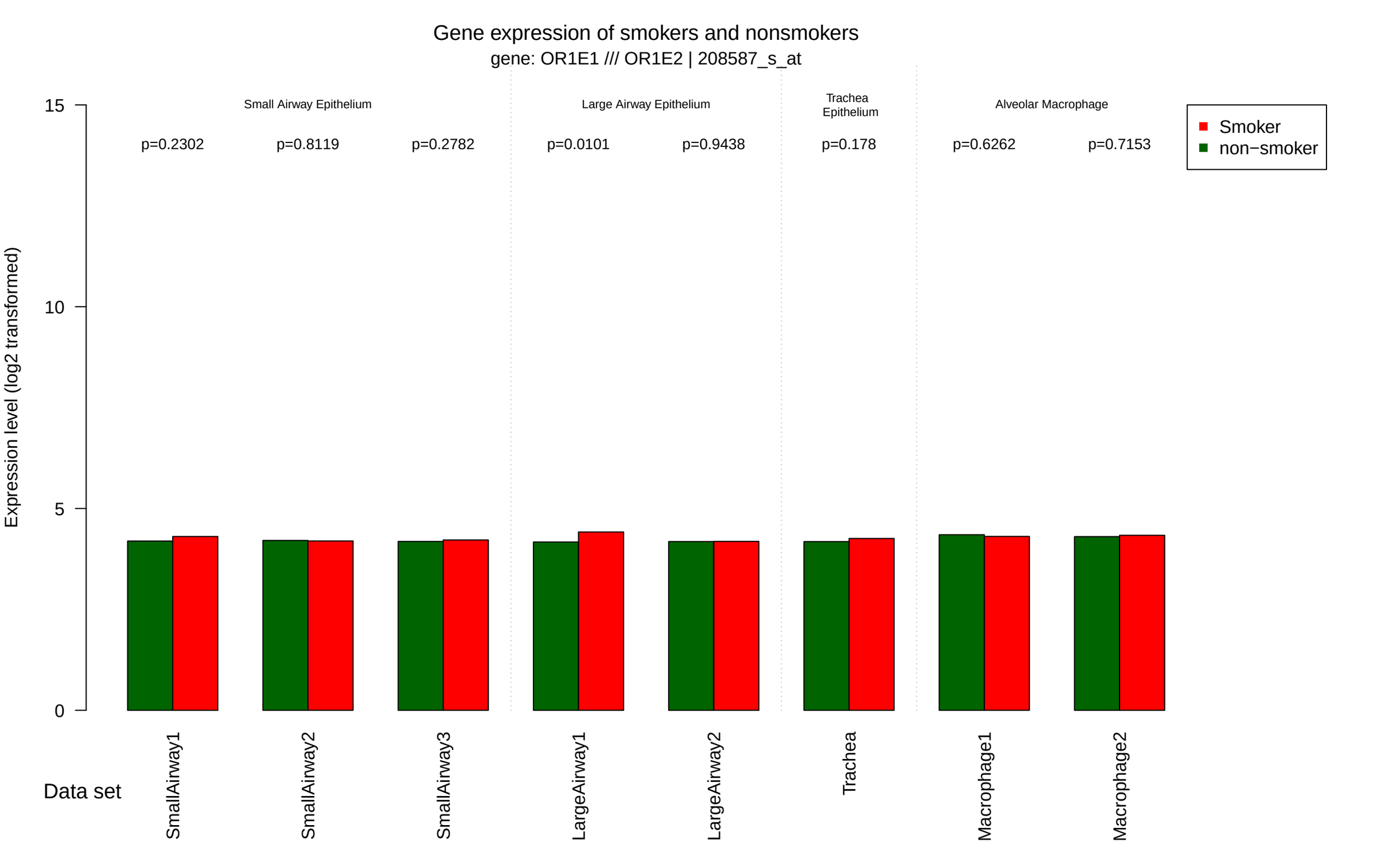
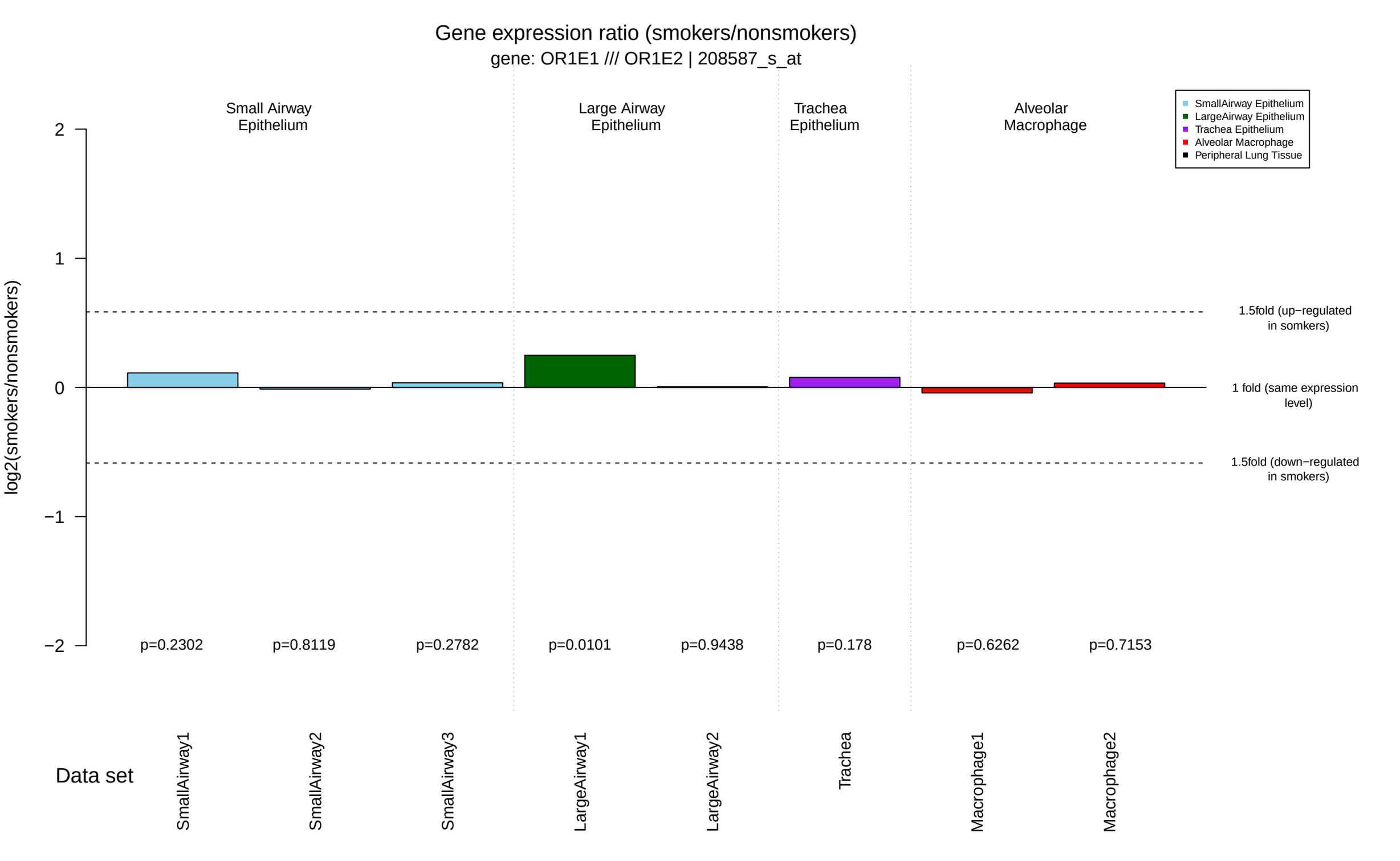
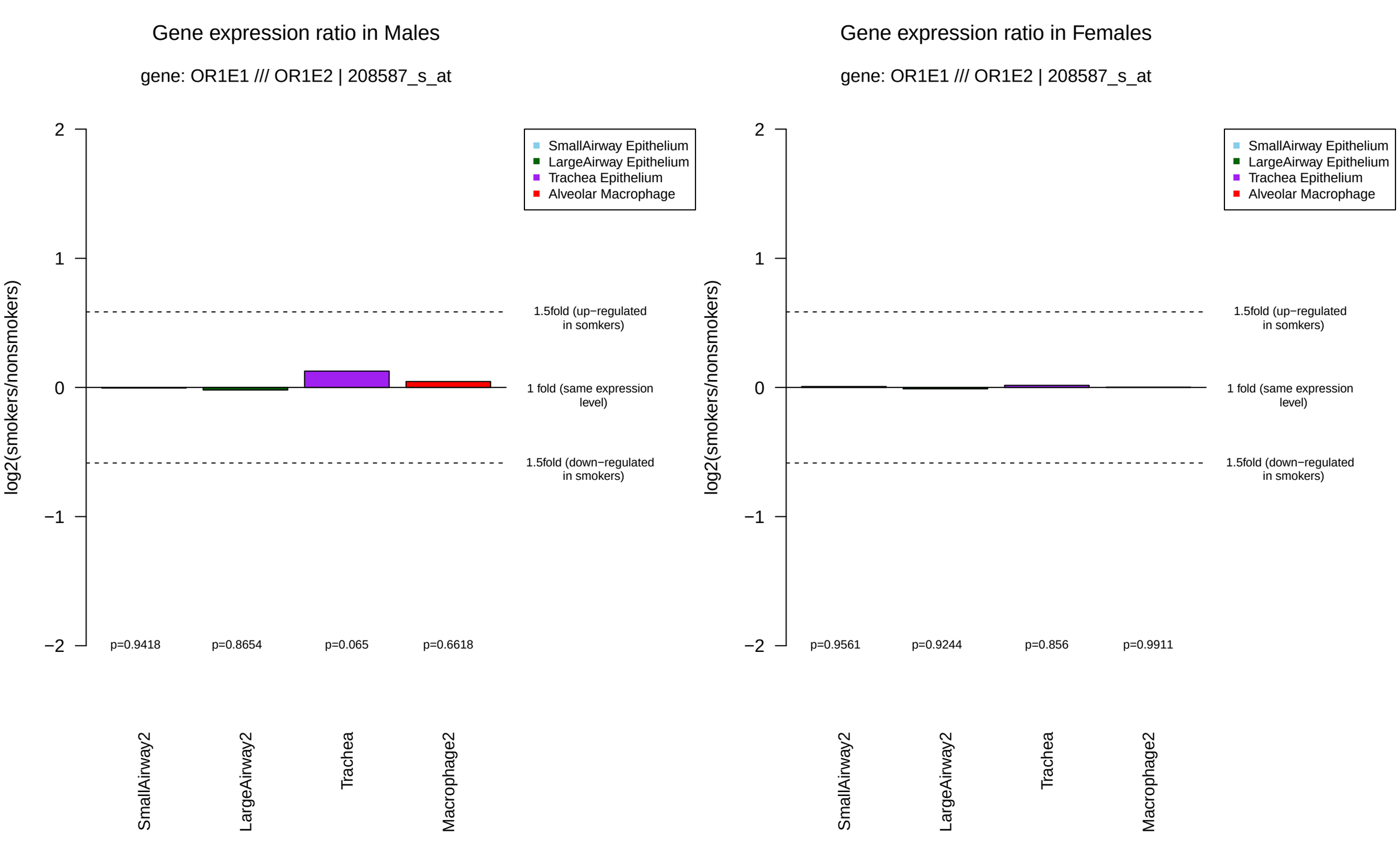
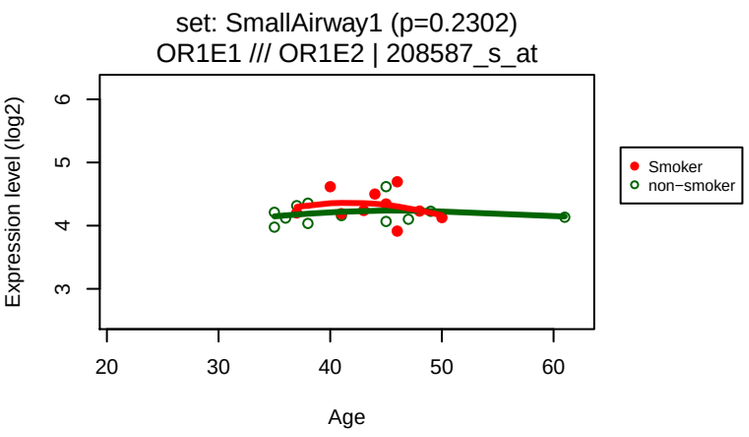
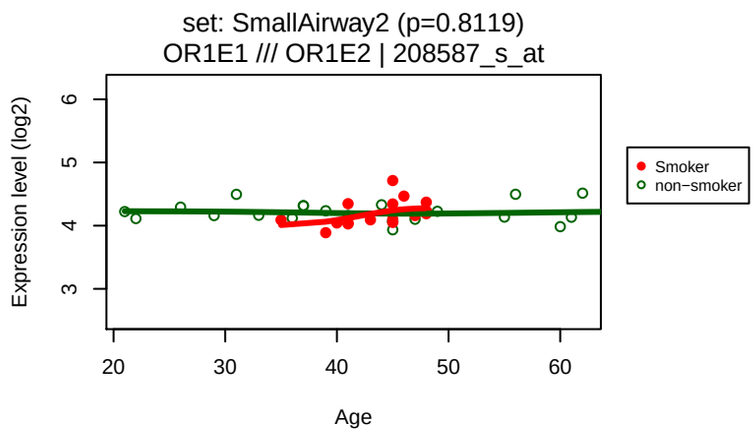
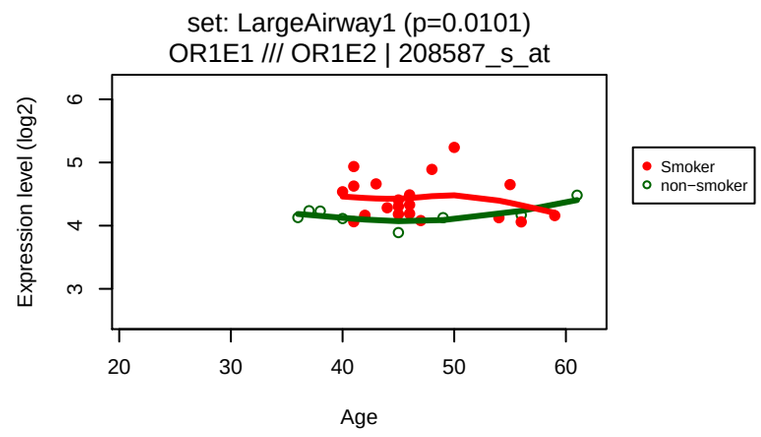

Gene expression of smokers and nonsmokers
gene: OR1E1 /// OR1E2 | 208587_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2302
p=0.8119
p=0.2782
p=0.0101
p=0.9438
p=0.178
p=0.6262
p=0.7153
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

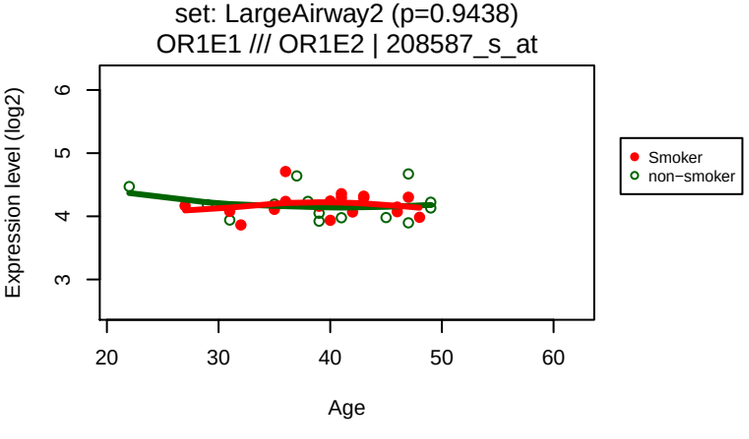

Gene expression ratio (smokers/nonsmokers)
gene: OR1E1 /// OR1E2 | 208587_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2302
p=0.8119
p=0.2782
p=0.0101
p=0.9438
p=0.178
p=0.6262
p=0.7153
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

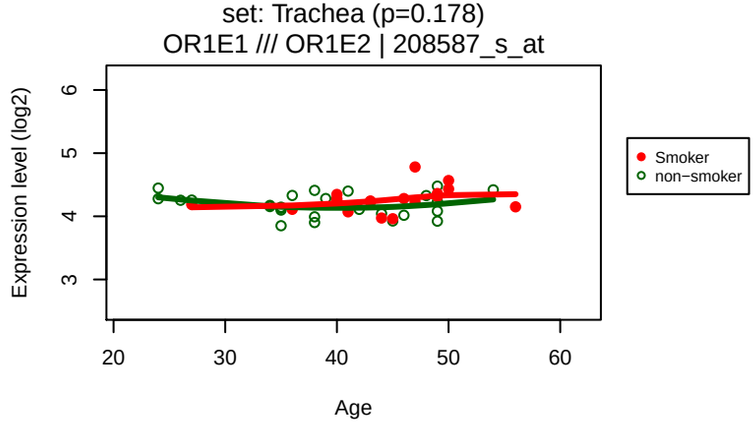

Gene expression ratio in Males
gene: OR1E1 /// OR1E2 | 208587_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9418
p=0.8654
p=0.065
p=0.6618
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR1E1 /// OR1E2 | 208587_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9561
p=0.9244
p=0.856
p=0.9911
SmallAirway2
LargeAirway2
Trachea
Macrophage2

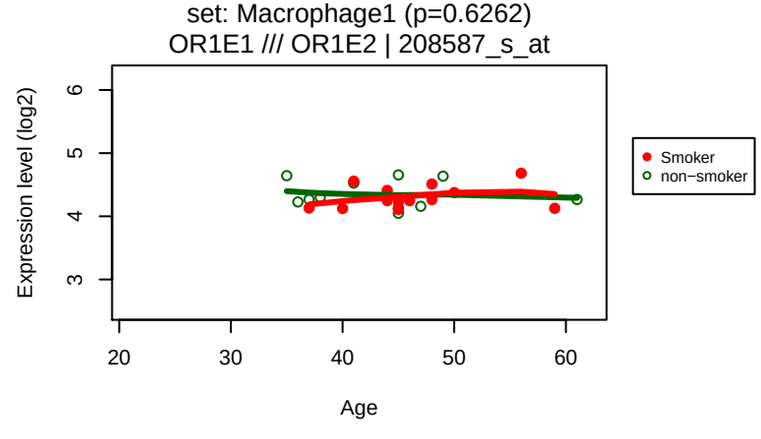

set: SmallAirway1 (p=0.2302)
OR1E1 /// OR1E2 | 208587_s_at
Expression level (log2)
Age
Smoker
non−smoker

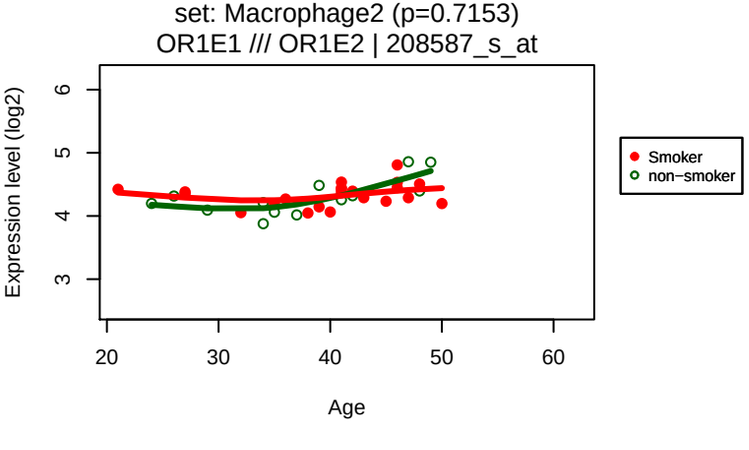

set: SmallAirway2 (p=0.8119)
OR1E1 /// OR1E2 | 208587_s_at
Expression level (log2)
Age
Smoker
non−smoker

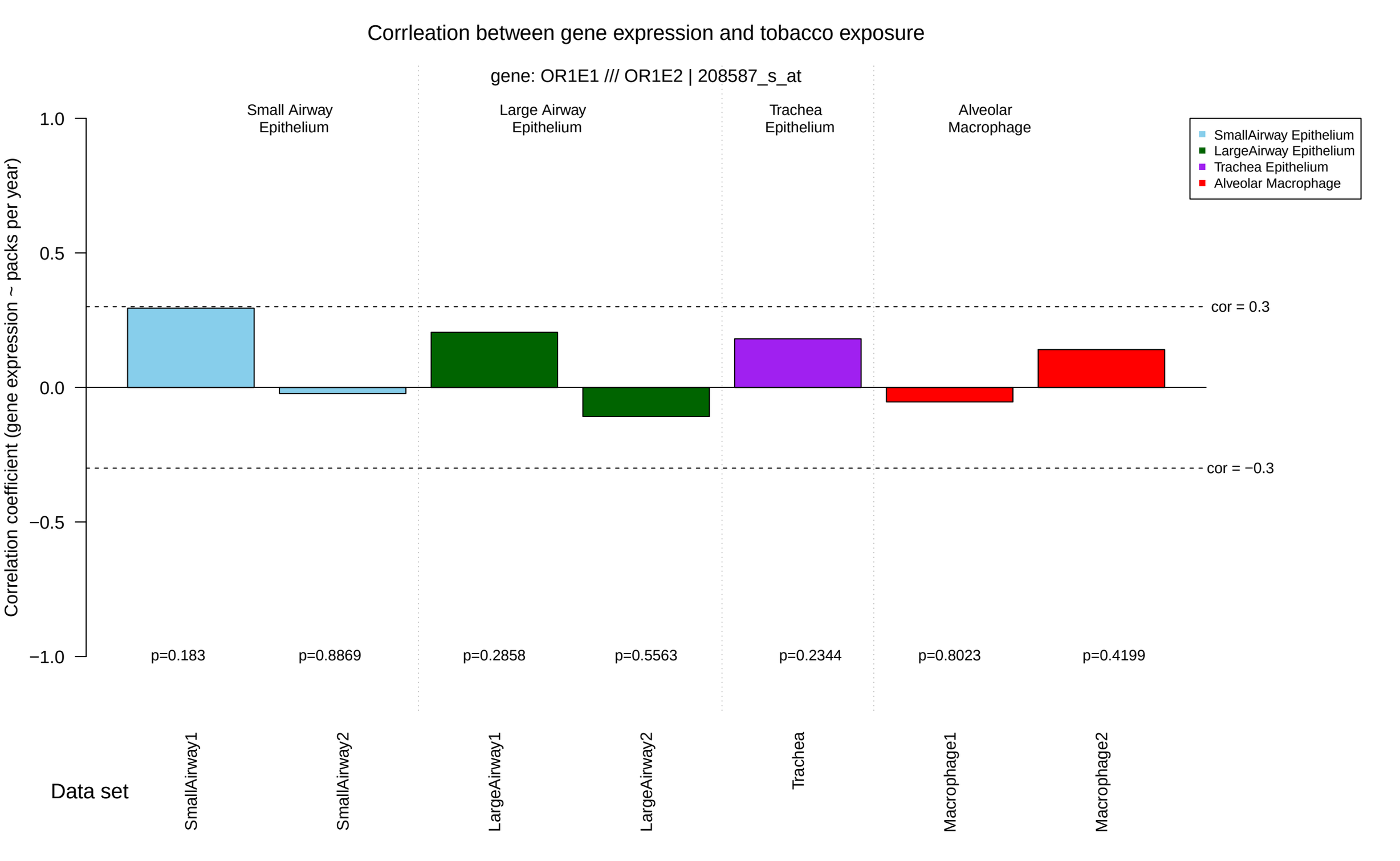

set: LargeAirway1 (p=0.0101)
OR1E1 /// OR1E2 | 208587_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9438)
OR1E1 /// OR1E2 | 208587_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.178)
OR1E1 /// OR1E2 | 208587_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6262)
OR1E1 /// OR1E2 | 208587_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7153)
OR1E1 /// OR1E2 | 208587_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR1E1 /// OR1E2 | 208587_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.183
p=0.8869
p=0.2858
p=0.5563
p=0.2344
p=0.8023
p=0.4199
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2